RPractical_Virgo

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Part A - Basic Operations

```
v <- seq(c(5, 10, 15, 20, 25, 30, 35, 40, 45, 50))
length(v)
```

A1

```
## [1] 10
```

```
sum(v)
```

```
## [1] 55
```

```
mean(v)
```

```
## [1] 5.5
```

```
w <- rep(c(1,2,3), times = 4)
z <- rep(c(1,2,3), each = 4)

print(w)
```

A2

```
##  [1] 1 2 3 1 2 3 1 2 3 1 2 3
```

```
print(z)
```

```
##  [1] 1 1 1 1 2 2 2 2 3 3 3 3
```

```
17 %% 5
```

A3

```
## [1] 2
```

```
17 %/% 5
```

```
## [1] 3
```

```
2^10
```

```
## [1] 1024
```

```
sqrt(144)
```

```
## [1] 12
```

```r
round(pi, 3)
```

```
## [1] 3.142
```

```r
a <- c(2, 4, 6, 8)
b <- c(1, 3, 5, 7)

print(a + b)
```

A4

```
## [1]  3  7 11 15
```

```r
print(a * b)
```

```
## [1]  2 12 30 56
```

```r
print(a > b)
```

```
## [1] TRUE TRUE TRUE TRUE
```

```r
print(sum(a*b))
```

```
## [1] 100
```

```r
paste("Student", 1:3, sep = "-")
```

A5

```
## [1] "Student-1" "Student-2" "Student-3"
```

```r
class (42)
```

```
## [1] "numeric"
```

```r
class (42L)
```

```
## [1] "integer"
```

```r
class("42")
```

```
## [1] "character"
```

```r
class (41i)
```

```
## [1] "complex"
```

```r
class (TRUE)
```

```
## [1] "logical"
```

Part B: Data Frames and Conditional Statements

```r
students <- data.frame(
id = 1:8,
name = c("Alice", "Bruno", "Chen", "Dana", "Elif", "Farid", "Grace", "Hiro"),
sex = c("F", "M", "M", "F", "F", "M", "F", "M"),
section = c("A", "B", "A", "C", "B", "C", "A", "B"),
```

```
midterm = c(78, 85, 92, 67, 74, 88, 95, 59),
final = c(82, 79, 96, 71, 80, 91, 93, 65),
attendance = c(95, 88, 100, 72, 90, 85, 98, 60)
)

students
```

B1

```
##   id  name sex section midterm final attendance
## 1  1 Alice   F       A      78    82         95
## 2  2 Bruno   M       B      85    79         88
## 3  3  Chen   M       A      92    96        100
## 4  4  Dana   F       C      67    71         72
## 5  5  Elif   F       B      74    80         90
## 6  6 Farid   M       C      88    91         85
## 7  7 Grace   F       A      95    93         98
## 8  8  Hiro   M       B      59    65         60
```

```
str(students)
```

B2

```
## 'data.frame':    8 obs. of  7 variables:
##  $ id        : int  1 2 3 4 5 6 7 8
##  $ name      : chr  "Alice" "Bruno" "Chen" "Dana" ...
##  $ sex       : chr  "F" "M" "M" "F" ...
##  $ section   : chr  "A" "B" "A" "C" ...
##  $ midterm   : num  78 85 92 67 74 88 95 59
##  $ final     : num  82 79 96 71 80 91 93 65
##  $ attendance: num  95 88 100 72 90 85 98 60
```

```
dim(students)
```

```
## [1] 8 7
```

```
names(students)
```

```
## [1] "id"         "name"       "sex"        "section"    "midterm"
## [6] "final"      "attendance"
```

```
summary(students)
```

```
##        id              name               sex             section     
##  Min.   :1.00   Length   :8   Length   :8   Length   :8   Min.   :59.00
##  1st Qu.:2.75   N.unique :8   N.unique :2   N.unique :3   1st Qu.:72.25
##  Median :4.50   N.blank  :0   N.blank  :0   N.blank  :0   Median :81.50
##  Mean   :4.50   Min.nchar:4   Min.nchar:1   Min.nchar:1   Mean   :79.75
##  3rd Qu.:6.25   Max.nchar:5   Max.nchar:1   Max.nchar:1   3rd Qu.:89.00
##  Max.   :8.00                                            Max.   :95.00
##      final         attendance
##  Min.   :65.00   Min.   : 60.00
##  1st Qu.:77.00   1st Qu.: 81.75
##  Median :81.00   Median : 89.00
##  Mean   :82.12   Mean   : 86.00
##  3rd Qu.:91.50   3rd Qu.: 95.75
##  Max.   :96.00   Max.   :100.00
```

```r
students$average <-  round((students$midterm + students$final) / 2, 1)

students
```

B3

```
##   id  name sex section midterm final attendance average
## 1  1 Alice   F       A      78    82         95    80.0
## 2  2 Bruno   M       B      85    79         88    82.0
## 3  3  Chen   M       A      92    96        100    94.0
## 4  4  Dana   F       C      67    71         72    69.0
## 5  5  Elif   F       B      74    80         90    77.0
## 6  6 Farid   M       C      88    91         85    89.5
## 7  7 Grace   F       A      95    93         98    94.0
## 8  8  Hiro   M       B      59    65         60    62.0
```

```r
students$passed <- students$average >= 70 & students$attendance >= 75

students$passed
```

B4

```
## [1]  TRUE  TRUE  TRUE FALSE  TRUE  TRUE  TRUE FALSE
```

```r
students$standing <- ifelse(students$average >= 90, "Honors", ifelse(students$average >= 75, "Good", "N

print(students[, c("name", "average", "standing")])
```

B5

```
##    name average      standing
## 1 Alice    80.0          Good
## 2 Bruno    82.0          Good
## 3  Chen    94.0        Honors
## 4  Dana    69.0 Needs Support
## 5  Elif    77.0          Good
## 6 Farid    89.5          Good
## 7 Grace    94.0        Honors
## 8  Hiro    62.0 Needs Support
```

```r
classify <- function(score) {
  if (score >= 90){
    "Distinction"
  } else if (score >= 73){
    "Merit"
  } else if (score >= 55){
    "Pass"
  } else{
    "Fail"
  }

}
```

```r
result <- sapply(c(90, 73, 55, 40), classify)
result
```

B6

```
## [1] "Distinction" "Merit"       "Pass"        "Fail"
```

Part C: rbind() and cbind()

```r
new_students <- data.frame(
  id         = c(9, 10),
  name       = c("Ines", "Jamal"),
  sex        = c("F", "M"),
  section    = c("C", "A"),
  midterm    = c(81, 70),
  final      = c(84, 68),
  attendance = c(92, 80),
  stringsAsFactors = FALSE
)

new_students$average  <- round((new_students$midterm + new_students$final) / 2, 1)

new_students$passed   <- new_students$average >= 70 & new_students$attendance >= 75

new_students$standing <- ifelse(new_students$average >= 90, "Honors", ifelse(new_students$average >= 75

students_all <- rbind(students, new_students)
students_all
```

C1

```
##    id  name sex section midterm final attendance average passed      standing
## 1   1 Alice   F       A      78    82         95    80.0   TRUE          Good
## 2   2 Bruno   M       B      85    79         88    82.0   TRUE          Good
## 3   3  Chen   M       A      92    96        100    94.0   TRUE        Honors
## 4   4  Dana   F       C      67    71         72    69.0  FALSE Needs Support
## 5   5  Elif   F       B      74    80         90    77.0   TRUE          Good
## 6   6 Farid   M       C      88    91         85    89.5   TRUE          Good
## 7   7 Grace   F       A      95    93         98    94.0   TRUE        Honors
## 8   8  Hiro   M       B      59    65         60    62.0  FALSE Needs Support
## 9   9  Ines   F       C      81    84         92    82.5   TRUE          Good
## 10 10 Jamal   M       A      70    68         80    69.0  FALSE Needs Support
```

```r
quiz <- c(8, 7, 10, 6, 7, 9, 10, 5, 8, 6)
students_all <- cbind(students_all, quiz = quiz)
class(students_all)
```

C2

```
## [1] "data.frame"
```

```r
m <- cbind(midterm = students_all$midterm, final = students_all$final)

m <- rbind(m, mean = colMeans(m))
print(m)
```

C3

```
##      midterm final
##         78.0  82.0
##         85.0  79.0
##         92.0  96.0
##         67.0  71.0
##         74.0  80.0
##         88.0  91.0
##         95.0  93.0
##         59.0  65.0
##         81.0  84.0
##         70.0  68.0
## mean    78.9  80.9
```

```r
# rbind() R pairs up the columns by name. If the column names don't match, R gets gives you the error "
```

C4

Part D: Subsetting

```r
students_all$name
```

D1

```
##  [1] "Alice" "Bruno" "Chen"  "Dana"  "Elif"  "Farid" "Grace" "Hiro"  "Ines"
## [10] "Jamal"
```

```r
students_all[["name"]]
```

```
##  [1] "Alice" "Bruno" "Chen"  "Dana"  "Elif"  "Farid" "Grace" "Hiro"  "Ines"
## [10] "Jamal"
```

```r
students_all[, "name"]
```

```
##  [1] "Alice" "Bruno" "Chen"  "Dana"  "Elif"  "Farid" "Grace" "Hiro"  "Ines"
## [10] "Jamal"
```

```r
students_all[2:5, c("name", "midterm", "final")]
```

D2

```
##    name midterm final
## 2 Bruno      85    79
## 3  Chen      92    96
## 4  Dana      67    71
## 5  Elif      74    80
```

```
students_all[students_all$attendance > 90, ]
```

D3

```
##   id  name sex section midterm final attendance average passed standing quiz
## 1  1 Alice   F       A      78    82         95    80.0   TRUE     Good    8
## 3  3  Chen   M       A      92    96        100    94.0   TRUE   Honors   10
## 7  7 Grace   F       A      95    93         98    94.0   TRUE   Honors   10
## 9  9  Ines   F       C      81    84         92    82.5   TRUE     Good    8
```

```
subset(students_all, sex == "F" & section %in% c("A", "C") & final >= 75, select = c(name, section, fina
```

D4

```
##    name section final
## 1 Alice       A    82
## 7 Grace       A    93
## 9  Ines       C    84
```

```
students_all$name[which.max(students_all$average)]
```

D5

```
## [1] "Chen"
```

```
students_all$name[which.min(students_all$average)]
```

```
## [1] "Hiro"
```

```
sorted <- students_all[order(students_all$average, decreasing = TRUE), ]

head(sorted, 3)
```

D6

```
##   id  name sex section midterm final attendance average passed standing quiz
## 3  3  Chen   M       A      92    96        100    94.0   TRUE   Honors   10
## 7  7 Grace   F       A      95    93         98    94.0   TRUE   Honors   10
## 6  6 Farid   M       C      88    91         85    89.5   TRUE     Good    9
```

Part E: Basic Factors

```
students_all$section <- factor(students_all$section)
students_all$sex <- factor(students_all$sex)

levels(students_all$section)
```

E1

```
## [1] "A" "B" "C"
```

```
levels(students_all$sex)
```

```
## [1] "F" "M"
```

```r
table(students_all$sex)
```

E2

```
##
## F M
## 5 5
```

```r
table(students_all$section)
```

```
##
## A B C
## 4 3 3
```

```r
table(students_all$sex, students_all$section)
```

```
##
##     A B C
##   F 2 1 2
##   M 2 2 1
```

```r
# The table shows sex on the left and section on top. Each box counts the students in that group. For e
```

```r
as.integer(students_all$sex)
```

E3

```
##  [1] 1 2 2 1 1 2 1 2 1 2
```

```r
students_all$sex_label <- factor(students_all$sex, levels = c("F", "M"), labels = c("Female", "Male"))

levels(students_all$sex_label)
```

E4

```
## [1] "Female" "Male"
```

```r
summary(students_all$sex_label)
```

E5

```
## Female   Male
##      5      5
```

```r
mean(students_all$average[students_all$sex_label == "Female"])
```

```
## [1] 80.5
```

```r
mean(students_all$average[students_all$sex_label == "Male"])
```

```
## [1] 79.3
```